

Music Bot Documentation

Discord multi-worker music bot: `main.py` controller, `worker.py` workers, `db.py` storage and `appsettings.py` settings.

This document explains every function, class and method of the source files, the settings you can change, the rules for who can control playback, the saved-songs ("algo") and autoplay system, and the messages the controller and the workers exchange.

Contents

1. [Architecture overview](#)
2. [Configuration](#)
3. [Commands](#)
4. [Who can control playback](#)
5. [Saved songs \(algo\) and autoplay](#)
6. [main.py reference \(controller\)](#)
7. [worker.py reference \(worker\)](#)
8. [db.py and appsettings.py reference](#)
9. [IPC operations and state events](#)
10. [Tunable constants and files](#)
11. [Graceful restart](#)
12. [Deployment and service](#)
13. [Lyrics](#)
14. [Playback smoothness](#)

1. Architecture overview

The system has four parts that run together on the server:

Part	File / service	Job
Controller	<code>main.py</code> , started by <code>launcher.py</code> (systemd unit <code>SERVER_musicbots</code>)	Logs in as the main bot, registers the slash commands for every server, enforces the whitelist, reads the dev <code>prefix</code> commands, starts and watches the workers, decides which worker serves which request, and relays commands to them.
Workers	<code>worker.py</code> , started once per worker token with <code>--index N</code>	Each copy is a separate Discord bot in its own process. Worker N uses <code>tokens[N]</code> . A worker keeps one session per server , so it can play in one voice channel in every server at the same time. It searches, resolves streams, plays, records listens and runs autoplay.
Storage	<code>db.py</code> → <code>database/</code>	SQLite files. <code>database/server/servers.db</code> holds per-server settings; <code>database/algo/<server_id>.db</code> holds every user's saved songs for that server.
Lavalink	<code>localhost:2333</code> (unit <code>SERVER_lavalink</code>)	The audio server, with the youtube-plugin and lavasrc-plugin. Workers connect to it with wavelink. YouTube streams are resolved by yt-dlp first and given to Lavalink as plain URLs.

How a play request flows

1. A user runs `/play` in Discord. The controller checks the server is whitelisted and the control channel (`_guard`).
2. The controller takes the pick lock, asks every worker for its real state (`sync`), and chooses a worker and voice channel (`_pick_for_play`). A new worker is *reserved* for that user before the lock is released, so two simultaneous `/play` s never get the same worker.
3. The controller sends `search_and_play` to that worker over its Unix socket `.worker<N>.sock` , including the user id.
4. The worker creates a session for that server (the user becomes its **controller**), searches (Spotify for clean metadata, YouTube Music for candidates), scores the candidates, connects to the voice channel and plays the best one. YouTube tracks are resolved to a direct stream URL with yt-dlp. The worker sets the voice channel status to `🎵Playing - <title>` .
5. The worker replies with the track details and a session snapshot; the controller turns them into an embed. Later track changes are announced by the worker itself in the text channel.
6. Whenever a session starts, changes or ends, the worker sends a `state` event to the controller over `.main.sock` , so the controller's view of who is busy stays correct even when a worker leaves on its own (idle timeout, kicked, failures).

Discord users never talk to the workers directly. All user input goes through the controller.

2. Configuration

There are three places settings live:

Where	Scope	Edited by
<code>server_settings.json</code>	Whole bot (every server)	You, on the server. Contains secrets, git-ignored.
<code>.env</code>	Whole bot	You, on the server. <code>GROQ_API_KEY</code> for the title filters and AI autoplay fallback.
<code>database/server/servers.db</code>	One server	Server admins, with <code>/settings</code> in Discord.

server_settings.json

Copy `server_settings.json.example` and fill it in. Behaviour keys are re-read automatically when the file changes (no restart needed). `tokens` and `lavalink` are only read at start-up, so changing them needs a restart.

Key	Default	Meaning
<code>tokens</code>	— (required)	<code>tokens[0]</code> is the main controller bot, <code>tokens[1]</code> , <code>tokens[2]</code> , ... are the workers. One worker process is started per extra token. At least 2 tokens are required.
<code>lavalink.uri</code>	— (required)	Lavalink address, e.g. <code>http://localhost:2333</code> .
<code>lavalink.password</code>	—	Lavalink password.
<code>dev_ids</code>	<code>[]</code>	Discord user ids of the bot developers. They can use the <code>prefix</code> commands in any server and always control playback.
<code>whitelist_enabled</code>	<code>true</code>	When true, the bot only works in servers listed in <code>whitelist</code> .
<code>whitelist</code>	<code>[]</code>	Server ids the bot may be in.
<code>leave_message</code>	"This server is not currently whitelisted, bot will not function"	Posted (with the server owner pinged) before the bot leaves a non-whitelisted server.
<code>algo_max_kb</code>	<code>1024</code>	Storage cap for one user's saved songs on one server, in KB. There is no song-count limit; a saved song takes about 100 bytes, so 1 MB is roughly 10,000 songs.
<code>autoplay_seed_count</code>	<code>5</code>	How many songs autoplay looks at when choosing what to play next.
<code>lyric_safe</code>	<code>true</code>	Lyrics layout (both are an embed with the previous, current and next block divided by solid lines). <code>true</code> : two lyric lines per block, so it changes about half as often. <code>false</code> : one lyric line per block. Applies from the next song.
<code>lyrics_delay</code>	<code>-0.2</code>	Lyrics timing in seconds (decimals allowed). Negative shows lines earlier, positive later. It's added to the built-in 0.4 s head start (<code>LYRICS_EDIT_LEAD</code>) that covers the time an edit takes to show in Discord. Takes effect within a second, no restart.

Per-server settings (/settings)

Stored in `database/server/servers.db`, one row per server. Change them with the `/settings` panel (section 3). `get-env` shows them.

Key	Default	Meaning
<code>CC_ID</code>	not set	Control channel. Slash commands only work there and workers post their now-playing messages there. <code>/settings</code> itself works in any channel, so a wrong value cannot lock admins out.
<code>VCW<n></code>	not set	Fixed voice channel of worker n. A worker with this setting always joins that channel and is not borrowed for other channels. A worker without it joins the voice channel of the person who used the command.
<code>SRC</code>	<code>sp</code>	Default search source for <code>/play</code> and <code>/playlist</code> : <code>sp</code> Spotify, <code>yt</code> YouTube Music, <code>sc</code> SoundCloud. The command's <code>source</code> option overrides it.
<code>P_EPH</code>	false	<code>/play</code> replies private (only the person who ran it sees them).
<code>PL_EPH</code>	false	<code>/playlist</code> replies private.
<code>AP_EPH</code>	same as <code>P_EPH</code>	<code>/autoplay</code> replies private. Until set on its own in <code>/settings</code> , it follows <code>/play</code> .
<code>HC_EPH</code>	false	<code>/hctest</code> replies private.
<code>CC_EPH</code>	false	<code>/control</code> panel and its button feedback private.
<code>S_EPH</code>	false	<code>/stop</code> replies private.

When playback is started with a private command (`P_EPH`, `PL_EPH` or `AP_EPH` on), the worker also stops posting "Now Playing" messages on track changes, and the "Playback failed" notice. Those are normal channel messages, which Discord can't make private. The current song still shows in the voice channel status and in `/control`. The setting follows the most recent play command on that worker.

Discord permissions the bots need

- **Main bot:** Message Content Intent enabled in the Developer Portal (for the `prefix` commands). Send Messages in the server's first channel (for the whitelist notice).
- **Lavalink:** the **LavaDSPX** plugin for the Normalize button (`com.github.Devoxin:LavaDSPX-Plugin:0.0.5`, repository <https://jitpack.io>, tested with Lavalink 4.2.2). Without it, Normalize replies with an error and everything else works.
- **Workers:** Connect and Speak in voice channels, Send Messages and Embed Links in the control channel (now-playing and lyrics messages; they also delete their own lyrics messages), and **Set Voice Channel Status** (to show the song title on the channel; without it playback still works and a log line notes the missing permission). **Manage Channels** only if a voice region is set in `/settings` (without it the region isn't changed and a log line says so).
- Every server needs the main bot **and** all worker bots invited.

3. Commands

Slash commands

Command	Who	What it does
<code>/play query [source] [worker]</code>	Anyone (subject to section 4)	Plays a song by name or URL, or queues it if your worker is already playing.
<code>/playlist query [source] [worker]</code>	Anyone (subject to section 4)	Plays or queues a whole playlist (Spotify, YouTube, SoundCloud, Apple Music).
<code>/autoplay [worker]</code>	Anyone with saved songs	Starts music picked from your saved songs on this server, without a query. If you already control a playing worker, it switches that worker's autoplay to your saved songs.
<code>/stop [worker]</code>	Controller (or anyone in "All" mode), devs, Manage Server	Stops the worker and makes it leave.
<code>/control [worker] [no_eph]</code>	Controller (or anyone in "All" mode), devs, Manage Server	Opens the playback control panel. <code>no_eph:true</code> makes the session public : its panel and every reply about it (<code>/play</code> , <code>/playlist</code> , <code>/autoplay</code> , <code>/stop</code> , <code>/control</code> , panel buttons) from people in that voice channel are visible to the channel, whatever the private-reply settings say. People in other voice channels aren't affected. Control is unchanged: in Me mode others can see the panel but not use it. <code>no_eph:false</code> turns it off.
<code>/hctest</code>	Anyone	Health check of the controller and every worker.
<code>/settings</code>	Manage Server or dev	Opens the per-server settings panel (only you can use it).
<code>/reset-algo</code>	Anyone (only themselves)	Opens your algo menu: all your saved songs on this server (25 per page, best first). Select songs to remove (🗑 Remove selected) or 🔄 Reset all; both ask for confirmation, and the remove confirmation lists the exact songs. Selections are by song, not position. There is no way to see or change someone else's. Works in any channel; always private.
<code>/purge [limit]</code>	Manage Messages or dev	Deletes bot messages among the last <code>limit</code> (1–100, default 20) messages in the channel.

The `worker` option forces a specific worker (0 = automatic). It is needed to use a worker fixed to a channel you are not in.

Control panel (/control)

Row	Contents
0	⏮ 10s, ⏮ 5s, ⏭ 5s, ⏭ 10s, 🔁 Loop

Row	Contents
1	Backward (restart track), Pause / Resume (shows which one applies), Stop, Skip, Mute (hides now-playing messages)

The panel **updates itself** (keeping your page and selection): - **right away** when the worker reports a change: the next song is picked (shown while it loads), a song starts, or the queue changes; - otherwise it checks every 4 s; - the header shows paused, muted, loop, autoplay, normalized and lyrics when they're on; - the bottom line shows " **Panel expires in: X min**", updated every minute. Discord only allows editing the panel for 15 minutes; it counts down to 14.

Panels that can't do anything any more **delete themselves**: - **The session ended** (Stop, idle timeout, kicked, ...): the panel shows "Stopped" or "Playback ended" for 5 s, then it's deleted. - **A new session started on that worker**: an old panel never controls the new session (each session has an id); it's deleted. - **A public panel expired** (14 min): deleted. A private panel is marked "Panel expired" instead. - **Restart**: open panels are deleted while the bot shuts down.

Before deleting for a session end, the panel confirms it with the worker, so a missed sync can't delete a working panel. Lyrics posts no message: the button just turns green. | 2 | Queue dropdown (25 per page; marks autoplay picks) | | 3 | Jump To, Remove, Clear selection, Autoplay, Control: Me / Control: All | | 4 | Normalize (evens out loud and quiet songs; default off; also applies to the shutdown message), Lyrics (live synced lyrics, see section 13), then page buttons for queues longer than one page |

The panel stops working after 15 minutes; run `/control` again.

Settings panel (/settings)

Row	Contents
0	Control channel picker (empty = any channel)
1	Worker picker (choose which worker's fixed channel to edit)
2	Fixed voice channel picker for that worker (empty = follow the user)
3	Private replies multi-select (P/PL/AP/HC/CC/S)
4	Source (cycles Spotify → YouTube Music → SoundCloud), Clear control channel, Clear worker channel, Voice region (opens a picker: Don't change / Automatic / a region; see section 14), Done

Changes save immediately. A voice channel already fixed to another worker is refused.

Dev prefix commands

Typed as a message wrapped in two backticks on each side. Only users in `dev_ids` can use them, in any server; other people get no reply.

Command	Effect
<code>restart</code>	Restarts all workers (music stops in every server).
<code>restart 2</code>	Restarts worker 2 only.
<code>restart full</code>	<code>systemctl restart SERVER_musicbots</code> (controller and all workers).
<code>shutdown</code>	Shuts down all workers; they stay down until <code>restart</code> .
<code>shutdown 2</code>	Shuts down worker 2 only.
<code>shutdown full</code>	<code>systemctl stop SERVER_musicbots</code> . Bring it back on the server with <code>systemctl start</code> .
<code>get-env</code> (or <code>show-env</code>)	Shows this server's settings, whitelist state and each worker's state here.

4. Who can control playback

- Whoever starts playback on a free worker (with `/play`, `/playlist` or `/autoplay`) becomes that session's **controller**. The session starts in **Me** mode.
- In **Me** mode, only the controller can play, queue, use the panel or stop. Anyone else in that voice channel is told the worker is controlled by the controller and to join another voice channel to get their own worker. They cannot even add songs to the queue.
- The controller (or a dev or Manage Server member) can switch the panel's  /  button to **All**, which lets everyone control that worker, and switch it back.
- **Devs** (`dev_ids`) and members with **Manage Server** bypass Me/All: they can always play, queue, use the panel and stop any session.
- `/control no_eph:true` makes a session's replies public (section 3) without changing who can control it: in Me mode the channel can watch the panel but not use it.
- When the session ends (stop, idle timeout, kicked, failures), the worker is free again and the next user to start playback on it becomes the new controller.

5. Saved songs (algo) and autoplay

Saving

- A song is saved for the user who **requested** it (autoplay picks are credited to the session's controller, or to the `/autoplay` user).
- A song counts as a **listen** once it finished or was played for at least 30 seconds (`PLAY_COUNT_MS`). The first listen adds it; later listens increase its play count.
- A **skip in under 30 seconds** adds a skip to a song that is already saved. Unsaved songs are not added by skips.
- $\text{Score} = \text{plays} - 0.5 \times \text{skips}$.
- There is no limit on the number of saved songs; only storage is capped (`algo_max_kb`). When a user goes over it, the songs with the **lowest score** are removed first, oldest last-play breaking ties ("most unused", not "oldest"). The song that was just played is never the one removed.
- Each server has its own file `database/algo/<server_id>.db`, holding every user's saved songs for that server.
- Songs are keyed by their YouTube video id when known (stable across Spotify/YouTube lookups), else their URL.

Search memory

For every `/play` search, the bot remembers what it led to *for you*: a listen (30 s or more) or a quick skip. Stored in the same per-server file (`user_queries`), using up to a quarter of your storage cap; the least recently used entries go first.

Your history steers search

When you `/play` something, your history on that server is a real part of the ranking (`_personal`):

Signal	Effect
You listened to a result for this same search before	+60, more with repeat listens
You skipped a result for this same search	-40
The exact upload is in your saved songs	+35, more with plays (a quick skip cancels a listen)
The same song (title + artist) from another upload	+30, more with plays
You know a song with this title by a different artist	-15 (probably not the one you mean)
Artist share of your listening	up to +25

- Song boosts fade with time since you last listened (half strength after about a month without listening).
- Uploads you listened to that **aren't in the search results** (e.g. found earlier via Spotify) are fetched and added as candidates, so "noite quente" can find the Flame Runner upload you played without typing the artist.
- Artist names are matched without YouTube channel suffixes ("Sxilwix - Topic" = "Sxilwix").
- **Your words still win:** if the search names an artist ("noite quente m22"), history can't pull results by other artists. A word counts as an artist name only when more results have it in the artist than in the title, so "morena" (a song title, even though one artist is called Illest Morena) still uses your history. Altered versions still need to be asked for.
- With strong personal evidence (+40 or more) for a result that matches the search, it beats the AI's generic "most popular official upload" pick.
- When Spotify is used, its match also prefers a song or artist you know (many songs share a title).

- `/reset-algo` deletes your saved songs and search memory on that server (with a confirm button; only ever your own).

Autoplay from `/control` (🎮 button)

- When the queue runs out, the worker picks songs related to the **last few songs played in this session** (`autoplay_seed_count`), whether they came from `/play` or a playlist.
- It queues **5** picks at a time (`AUTOPLAY_BATCH`), in the background while the last song is still playing, so there is no gap. Their source shows as **Autoplay (Algo)**.
- Songs you queue yourself always go **ahead** of autoplay picks.
- Turning autoplay off removes the queued autoplay picks.
- It needs at least one song played in the session first.

`/autoplay`

- Seeds from **your saved songs** instead of what was played. Better-scored songs are more likely to be used as seeds.
- Its familiar picks (see below) start with your saved songs themselves, drawn at random and weighted by plays.
- Errors if you have no saved songs yet on this server.

How picks are found

1. **YouTube Mix:** for each seed song, the worker loads YouTube's own radio mix (`watch?v=ID&list=RDID`); the first 30 songs are used (`AUTOPLAY_MIX_DEPTH`). Mixes are cached for the session, so later refills don't reload them.
2. All seeds' Mix songs go into **one pool**, filtered: nothing already played in the session or queued (by video id and cleaned-up title), no altered versions, nothing under 1 minute or over 10 minutes.
3. Each candidate is scored (`_autoplay_score`): - how related it is: earlier in a Mix and from a stronger/more recent seed scores higher; appearing in **several seeds' Mixes** is a strong bonus; - **your reaction to autoplay before:** picks you listened to score higher, picks you skipped within 30 s are never played again (stored in search memory under `~autoplay`); - your taste (`_personal`): artists you play, songs you know; artists you keep skipping are pushed down; - a little randomness for variety.
4. The pool is split into songs **new to you** (not in your algo — neither that upload nor another upload of the song, including re-uploads like "Artist - Song (Lyrics)") and songs **you know**.
5. **Each pick has a 70% chance to be a new song** (`AUTOPLAY_NEW_SHARE`), otherwise a song you know. New songs are still chosen from your taste: they come from the Mixes of your songs, songs that several of your songs lead to rank first, artists you keep skipping are pushed down, and artists you play count a little (`AUTOPLAY_NEW_TASTE_W`) — about half of new picks are by artists you already play, the rest are new artists. If one kind runs out, the other fills the slot, so a refill is always full.
6. Picks are queued, at most **2 per artist** per refill (`AUTOPLAY_PER_ARTIST`).
7. **AI fallback** (only if the Mixes don't give enough): Groq is asked once for 10 similar songs; unused ones are kept for later refills. They're looked up without another AI call per song, and only songs new to you are used.

For `/autoplay`, seeds are your saved songs, weighted by plays **and** how recently you listened.

6. main.py reference (controller)

Module level: `TOKENS` comes from `server_settings.json` through `appsettings.require_startup()` (the program exits with a clear message if tokens or Lavalink are missing). `EPH_KEYS` lists the private-reply keys and the command each controls. `SOURCES` maps `sp / yt / sc` to their names. `_workers` maps worker index to its `WorkerProcess`. `_pick_lock` serialises worker picking. `WORKER_SCRIPT` is `worker.py`; `MAIN_SOCKET` is `.main.sock`.

Per-server config helpers

def _gcfg(guild_id) Returns the server's settings dict (`cc_id`, `vcw`, `eph`, `src`), loading it from the server DB the first time and caching it in memory.

def _save_gcfg(guild_id) Writes the cached settings of a server back to the server DB.

def _cfg_cc(guild_id) Returns the server's control channel id, or None.

def _cfg_vcw(guild_id, index) Returns worker `index`'s fixed voice channel in that server, or None.

def _cfg_src(guild_id) Returns the server's default search source (`sp`, `yt` or `sc`), `sp` when unset or invalid.

def _eph(guild_id, key) Returns whether replies for a key such as `P_EPH` are private in that server. Unset keys return False (public).

def _migrate_legacy_settings() One-time import of the old single-server `settings.json`. Its `CC_ID`, `VCW` and private-reply flags are stored under its `G_ID` in the server DB (unless that server already has settings), then the file is renamed to `settings.json.migrated`.

Worker process handle

class WorkerProcess One worker subprocess. Besides the process handle and socket path it keeps the controller's view of that worker: `sessions` (server id → session snapshot: `channel_id`, `controller_id`, `mode`, `autoplay`, `text_channel_id`), pending reservations, the servers the worker bot is in (`guild_ids`), its bot user id and Lavalink state.

- **__init__(self, index)** — Stores the index, the socket path `.worker<index>.sock` and empty state.
- **session(self, guild_id)** — The worker's session in that server: the confirmed snapshot, else a reservation that hasn't expired (`PENDING_TTL`, 90 s), else None.
- **busy_in(self, guild_id)** — True when the worker has a session (or reservation) in that server.
- **reserve(self, guild_id, channel_id, user_id)** — Marks the worker as taken in that server by that user while a `/play` is in flight, before the worker confirms.
- **apply(self, guild_id, snap)** — Stores a session snapshot from the worker, or removes it when `snap` is None.
- **clear_state(self)** — Forgets all sessions and reservations (used when the worker restarts or dies).
- **start(self)** — Deletes a stale socket file, then spawns `python -u worker.py --index N` with its output piped to the controller, in its own process group. Starts `_stream_output` and `_reap`.
- **_stream_output(self)** — Prints each line of the worker's output prefixed with `[Worker N]`, so it shows in `journalctl`.
- **_reap(self)** — Waits for the process to exit in a thread so it doesn't become a zombie.
- **is_alive(self)** — True when the process exists and hasn't exited.
- **terminate(self, timeout=8.0)** — Stops the worker in three steps: ask it to run `shutdown_graceful` (up to 6 s), `SIGTERM` the process group and wait, then `SIGKILL`.
- **restart(self)** — Clears its state and shutdown flag, terminates it, waits 1 s and starts it again.

- `send(self, cmd, timeout=15.0)` — Sends one JSON command to the worker's socket and returns the JSON reply. Connecting is retried up to 3 times. Never raises: timeouts, empty replies and other failures become `{"status": "error", ...}`.

Worker state

`def _ordered_workers()` All workers sorted by index.

`def _worker_for_channel(guild_id, channel_id)` The worker that has a session in that voice channel of that server, or None.

`async def _resync_worker(w, timeout=4.0)` Asks a worker for its real state (`sync`) and replaces the controller's copy: servers it is in, bot user id, Lavalink state and all sessions. Returns False if the worker doesn't answer. The worker is the source of truth for busy/free.

`async def _resync_all()` Resyncs every worker at once (3 s timeout each).

`async def _handle_event(reader, writer)` Handles one `state` event from a worker (one JSON line) and applies the session snapshot it carries.

`async def _event_server()` Starts the Unix socket server on `.main.sock` that receives worker state events.

Embeds and permission helpers

`def _embed_from_response(resp)` Turns a worker reply into an embed using its `embed_type`: `playing` / `queued` (linked title, artist, thumbnail, duration, source, queue position) or `playing_playlist` / `queued_playlist` (name, track count, source, skipped local files, artwork). None when there is no `embed_type`.

`def _simple_embed(title, colour)` An embed with only a title and colour.

`def _err_embed(msg)` A red embed whose title is the message prefixed with .

`def _all_busy_embed(guild_id)` The red "No bots available" embed, counting busy workers among those in this server.

`def _is_dev(user)` True when the user is in `dev_ids`.

`def _has_perm(user, perm)` True when the member has a server permission such as `manage_guild`.

`def _allowed(session, user)` The control rule: True for devs, members with Manage Server, the session's controller, or anyone when the session is in All mode.

`def _session_public(interaction, worker_arg=0) / def _reply_eph(interaction, key, worker_arg=0)` Whether a reply is private: the server's setting for the command, unless the session it goes to (found from the user's voice channel or the `worker` option, using the cached state so it's decided within Discord's 3 seconds) was made public with `/control no_eph:true`.

`def _locked_msg(w, session)` The " Worker N is controlled by @X. Join another voice channel to get your own worker." text.

`def _message_embed(resp)` Turns a short worker reply (`stopped` , `paused` , `skipped` , `autoplay_on` , `mode_all` , `seeked` , ...) into a feedback embed using `_MESSAGE_EMBEDS`. None for unknown messages.

Main bot, whitelist and events

`class MainBot(discord.Client)` The controller's Discord client with the slash-command tree.

- `__init__(self)` — Default intents plus `message_content` (for the `prefix` commands).
- `setup_hook(self)` — Runs once at start-up. Logs in briefly with every worker token to clear any slash commands registered on the worker applications. (The main bot's commands are registered per server in `on_ready`.)

async def _enforce_whitelist(guild) Returns True for a whitelisted server. Otherwise it posts `leave_message` with the server owner pinged in the system channel (or the first text channel it can write in, so it never spams every channel), leaves the server and returns False.

async def on_ready() Logs the login and enforces the whitelist for every server the bot is in. On the first ready it registers the slash commands on each whitelisted server (`_sync_guild`) and removes any global copies, so nothing shows twice. Per-server commands appear in Discord immediately; global ones can take a while to reach Discord apps.

async def _sync_guild(guild) Registers all slash commands on one server.

async def on_guild_join(guild) Enforces the whitelist when the bot is added to a new server and registers the commands there right away.

Dev prefix commands

def _env_summary(guild) The `get-env` text: whitelist state, `CC_ID`, `SRC`, each worker's `VCW<n>` with its state here (busy, free or not in this server), and every private-reply flag.

async def on_message(message) Reacts only to messages wrapped in double backticks whose content is `restart`, `shutdown`, `get-env` or `show-env` (with an optional argument), and only from devs. Everyone else is ignored silently.

async def _systemctl(action) Runs `systemctl <action> SERVER_musicbots` without blocking. Change `SERVICE_NAME` if your unit has another name.

async def _delayed_systemctl(action, delay=2.0) Waits so the reply reaches Discord, then runs `_systemctl`.

async def _dev_power(message, cmd, arg) Body of `restart / shutdown`. `full` runs `systemctl`. No argument (or `all / 0`) targets every worker; a number targets one. Restart waits up to 15 s per worker for its socket to answer; shutdown marks workers so the watcher doesn't revive them. The reply is edited with a per-worker result.

Slash command helpers

async def _guard(interaction, during_restart=False) First step of the slash commands, before deferring so refusals are always private. Refuses commands outside a server, in a non-whitelisted server, or outside the control channel. While a restart lets songs finish, commands that start playback are refused; `/stop` and `/control` pass `during_restart=True` and keep working, so people can still stop or skip.

def _text_channel_id(interaction) Where the worker posts now-playing messages: the control channel when set, else the channel the command was used in.

def _user_vc_id(interaction) The voice channel the command author is in, or None.

async def _fail(interaction, msg, embed=None) Sends a private error follow-up.

async def _pick_for_play(interaction, worker_arg) Chooses `(worker, voice channel)` for `/play`, `/playlist` and `/autoplay`, or replies with an error and returns None. Runs under `_pick_lock` after `_resync_all`, and only considers workers that are in this server. - With the `worker` option: that worker. If it already has a session here, you must be allowed to control it; otherwise it joins its fixed channel or yours, unless another worker is already playing there. - When you are in a voice channel: the worker already there (if you are allowed to control it, otherwise the lock message), else a free worker fixed to your channel, else a free worker with no fixed channel, else "No bots available". - When you are in no voice channel: a free worker that has a fixed channel. - A newly chosen worker is reserved for you before the lock is released.

async def _pick_for_control(interaction, worker_arg, for_stop=False) Chooses `(worker, session)` for `/stop` and `/control` after a resync. With the `worker` option, that worker's session here. Otherwise the session in your voice channel, or the single session here that you can control. Refuses with the lock message when you are not allowed; with `for_stop`, Manage Server members are also allowed.

async def _dispatch(interaction, cmd, worker, eph=False, timeout=30.0) Sends a command to a worker and posts the result as a track/playlist embed, a short feedback embed or an error. Returns the worker's reply.

async def _start_playback(interaction, op, eph_key, worker_arg, timeout=30.0, **extra) Shared body of `/play`, `/playlist` and `/autoplay`: defers (private if the key says so), picks a worker, sends the operation with the user id and text channel, clears the reservation afterwards and stores the session snapshot from the reply.

Slash commands

async def slash_play(interaction, query, source=None, worker=0) `/play`. Uses `P_EPH`. Sends `search_and_play` with the chosen or server-default source.

async def slash_playlist(interaction, query, source=None, worker=0) `/playlist`. Uses `PL_EPH`. Sends `search_and_playlist`.

async def slash_autoplay(interaction, worker=0) `/autoplay`. Uses `AP_EPH`. Sends `autoplay_start` with a 60 s timeout (finding the first song can take a while).

async def slash_stop(interaction, worker=0) `/stop`. Uses `S_EPH`. Picks with `for_stop=True`, sends `stop`, and marks the session gone only when the stop succeeded.

async def slash_control(interaction, worker=0) `/control`. Uses `CC_EPH`. Picks the session, builds the panel and sends it.

async def slash_hctest(interaction) `/hctest`. Uses `HC_EPH`. Resyncs and pings every worker, then shows the controller status (latency, busy workers here, server count) and per worker: Discord and Lavalink state, whether it is busy here (with channel, controller and queue length), free or not in this server, how many servers it is active in, and its fixed channel.

async def slash_settings(interaction) `/settings`. Needs Manage Server or dev. Skips the control-channel check so admins can always fix it. Sends the private settings panel.

async def slash_purge(interaction, limit=20) `/purge`. Needs Manage Messages or dev. Deletes bot messages among the last `limit` messages and posts the count.

Control panel interface

def _truncate(text, n=80) Shortens text to n characters with an ellipsis.

class QueueSelect(discord.ui.Select) The queue dropdown, one page of up to 25 tracks. Choosing a track only selects it for Jump To / Remove. Options are keyed by the song's `qid`, so the selection stays on the same song even when the queue shifts; the number shown is its current position. - **__init__(self, view, queue, page, total_pages, current_title, selected_idx=None)** — Builds the options (🎵 for autoplay picks), an empty-queue placeholder and the placeholder text. - **callback(self, interaction)** — Stores the selection and refreshes the panel.

class JumpToButton — 📺 Jump To, disabled until a track is selected. `callback` sends `jump_to`, reports the new track and refreshes from page 0.

class RemoveButton — 🗑️ Remove, disabled until a track is selected. `callback` sends `remove_from_queue` and refreshes the same page.

class ClearSelectionButton — ✖️ Clear. `callback` refreshes with no selection.

class AutoplayButton — 🎵 Autoplay / 🟢 Autoplay ON (green). `callback` sends `toggle_autoplay`.

class ModeButton — 🔒 Control: Me / 🔒 Control: All (green). `callback` sends `set_mode` with the other mode. Only the controller or a dev can press it (checked in `ControlView.interaction_check`).

class PrevPageButton, **class NextPageButton** — Move one queue page back/forward (disabled at the ends).

class PageLabelButton — Disabled "Page x / y" label.

class ControlView(discord.ui.View) The panel (15 minute timeout). Rows as in section 3. - **__init__(self, worker, guild_id, queue, current_title, page=0, muted=False, loop=False, autoplay=False, mode="me", selected_qid=None, normalize=False, lyrics=False)** — Stores the state and adds the dropdown and dynamic buttons. Every version of the panel gets its own button ids (tag): each refresh puts a new view on the same message and retires the old one, and discord.py forgets a retired view's buttons by id for that message — with ids shared between versions (Stop, Autoplay, Control used to have fixed ones) that also unhooked the new view's buttons, so they did nothing after the first refresh. - **interaction_check(self, interaction)** — Runs before every button/dropdown. Refuses (privately) when the session has ended or the user isn't allowed (section 4). The mode button is controller/dev only; the Stop button is also allowed for Manage Server. - **_dispatch(self, interaction, op, timeout=15.0, **extra)** — Shared handler: sends the operation, stores any session snapshot, shows feedback, and refreshes the panel after skip, restart, pause/resume, mute, loop, autoplay, normalize, lyrics and mode changes. Returns the reply. - **rw10, rw5, ff5, ff10** — Seek -10/-5/+5/+10 s. - **btn_loop** — **toggle_loop**. - **btn_backward** — backward (restart the track). - **btn_pause** — **pause_resume**. - **btn_stop** — **stop**; only when it succeeds does it mark the session gone and disable the panel. - **btn_skip** — **skip** (45 s timeout, since it may need to fetch autoplay picks). - **btn_mute** — **toggle_mute** (hides now-playing messages; audio keeps playing).

class LyricsButton 🎧 Lyrics: sends **toggle_lyrics** with **on** set to the opposite of what the button shows. When turning on, it first registers a **PrivateLyrics** for this worker and server, so updates for private playback can't arrive before there is somewhere to show them.

class PrivateLyrics Lyrics for privately started playback. The worker sends each update as a **lyrics** event; this shows it as an ephemeral follow-up to the 🎧 click (one per song, deleted when the song ends). Updates carry a sequence number, so a late older update never overwrites a newer one. Discord allows follow-ups for 15 minutes after a click, so at **PANEL_LIFETIME** (14 min) it shows "Private lyrics expired — press 🎧 Lyrics again" and turns lyrics off on the worker. A newer 🎧 click retires the previous one.

async def _build_control_view(worker, guild_id, page=0, selected_idx=None) Fetches **get_queue**, stores the session snapshot, clamps the page, drops a selection that no longer exists, builds the view, colours the Mute/Loop buttons and builds the "Playback Controls — Worker N" embed (current track, flags, queue range, and who controls it).

class AlgoMenuView, class AlgoConfirmView, async def slash_reset_algo(interaction) /reset-algo: the algo menu (dropdown of saved songs by **track_key**, page buttons, 🗑️ Reset all, 🟡 Remove selected, Cancel) and its confirm step (Yes / Back). Only the person who opened it can use it. The Yes click is acknowledged right away (telling the workers to forget can take longer than Discord's 3-second limit), then the message is edited. After removing, the menu reopens with the updated list; workers drop their cached copy of the user's history (**forget_user**).

class PanelState, async def _panel_autorefresh(state) An open panel: its message, newest view, session id (**sid**), whether it's public, and a **wake** event. The panel is registered in **_open_panels** per (worker, server). A worker **panel** event sets **wake**, so the panel redraws within about 0.3 s; otherwise it checks every **PANEL_REFRESH** (4 s). It's edited only if what it shows changed (**view.sig**, which includes the minutes left, so the countdown updates once a minute). - **session()** — the panel's session, or None once it ended or another session (different **sid**) replaced it. - **ended()** — confirms that with the worker before anything is deleted; a worker that doesn't answer counts as "not sure". - **retire(note)** — shows the note for **PANEL_END_NOTE** (5 s), then deletes the message.

At 14 minutes a public panel is deleted, and a private one is marked "Panel expired".

ControlView.interaction_check refuses buttons on a panel whose session ended or was replaced. On shutdown, every open panel is deleted.

async def _refresh_control_panel(interaction, ctrl, page=0, selected_idx=None) Rebuilds the panel and edits the message in place. Errors are only logged.

Settings panel interface

def _settings_embed(guild, worker_idx, note=None) The settings embed: optional result note, control channel, each worker's fixed channel (marks the one being edited; workers not in the server are noted), default source and private-reply list.

def _channel_default(channel_id) The pre-selected value for a channel picker, or nothing.

class SettingsView(discord.ui.View) The /settings panel (10 minute timeout). Every change saves immediately and redraws the panel. - **__init__(self, invoker_id, guild, worker_idx)** — Builds the pickers with the current values selected. - **interaction_check(self, interaction)** — Only the person who ran /settings may use it. - **_redraw(self, interaction, note=None, worker_idx=None)** — Replaces the panel with a fresh one showing note. - **_on_cc** — Sets or clears the control channel. - **_on_worker** — Switches which worker's fixed channel is being edited. - **_on_vc** — Sets or clears that worker's fixed channel; refuses a channel already fixed to another worker; notes when the worker is playing elsewhere (applies from its next play). - **_on_eph** — Saves the private-reply flags. - **btn_source** — Cycles the default source Spotify → YouTube Music → SoundCloud. - **btn_clear_cc**, **btn_clear_vc** — Clear the control channel / the selected worker's fixed channel. - **btn_done** — Shows "Settings saved" and removes the controls.

Supervision and start-up

async def _watch_worker(w) Per-worker loop every 5 s: a dead worker (or one that doesn't answer a ping when the controller has no process handle) has its state cleared and is restarted after 3 s, unless it was shut down on purpose. Every 30 s it also resyncs the worker (backstop for lost state events) and health-checks it: a worker whose process is alive but that hasn't answered for **HUNG_AFTER** checks (2 minutes) is killed and restarted. Starts and restarts share **w.lock**, so the watchdog and a dev **restart** can never both start a process; a worker that doesn't answer is marked unavailable and skipped when picking a worker for **/play**.

async def _run(stop_event=None) Main coroutine. Migrates the old settings, starts the event server and the main bot, waits 5 s, creates one **WorkerProcess** per worker token, and for each either adopts an already-running worker (if its socket answers **sync**) or starts it, plus its watcher. Then waits for the stop signal and shuts everything down.

def _sig() SIGINT/SIGTERM handler in **--server** mode; sets the stop event.

7. worker.py reference (worker)

Module level: the worker index comes from `--index N` (falling back to a number at the end of the file name), its token is `tokens[N]` from `server_settings.json`, and the Lavalink address and password come from the same file. `GR0Q_API_KEY` is read from `.env`. All playback state lives in `Session` objects in `sessions` (server id → `Session`).

Session

class Session One server's playback on this worker.

Field	Meaning
<code>guild_id</code> , <code>channel_id</code>	Server and voice channel.
<code>controller_id</code> , <code>mode</code>	Who started playback; "me" or "all".
<code>text_channel</code> , <code>announce</code>	Where now-playing messages go; False when playback was started privately (no channel posts).
<code>queue</code>	List of <code>QueueItem</code> s: (track, source label, found-via, requester id) plus a stable <code>qid</code> .
<code>fallbacks</code> , <code>origin</code>	Alternate candidates per track; resolved track → original track.
<code>skipping</code> , <code>muted</code> , <code>loop</code>	Skip guard, hidden now-playing messages, repeat current track.
<code>autoplay</code> , <code>autoplay_source</code> , <code>autoplay_user</code> , <code>autoplay_task</code>	Autoplay on/off, "session" or "algo", whose saved songs, background refill task.
<code>fail_streak</code>	Failed tracks in a row.
<code>current_info</code> , <code>current_meta</code> , <code>current_requester</code>	Real title/artist of the playing track, its algo record, who requested it.
<code>idle_task</code> , <code>auto_paused</code>	Idle countdown; paused because the channel emptied.
<code>connecting</code> , <code>closing</code>	Guards so our own connect/disconnect isn't mistaken for being kicked.
<code>history</code>	The last 25 played tracks (autoplay seeds and duplicate filter).
<code>play_lock</code>	One song change at a time. wavelink only reports "playing" once a song is sent to Lavalink, so without it a <code>/play</code> landing while another song was still resolving or connecting started as well, and one song replaced the other. Taken by <code>_advance</code> and by the play commands from connecting until the song plays or is queued.

- **snapshot(self)** — The small dict sent to the controller: `channel_id`, `controller_id`, `mode`, `autoplay`, `text_channel_id`.

class QueueItem(tuple) A queued song: (track, source label, found-via, requester id) with a unique, never-reused `qid`. The control panel refers to songs by `qid`, not by position, because positions shift whenever a song ends or autoplay adds picks (which used to make Jump To / Remove hit the song *after* the one chosen).

def _queue_pos(sess, qid) Current position of the song with that `qid`, or None when it already played or was removed.

Groq filters

async def _is_altered(title) Asks the Groq model (`openai/gpt-oss-20b`) whether a title is an altered version (sped up, slowed, remix, cover, live, instrumental, ...). 4 s limit; errors count as clean.

async def _user_wants_altered(query) Asks whether the search text explicitly asks for an altered version, so a search for a remix isn't filtered. Errors count as "no". Queries containing an obvious word (sped up, slowed, reverb, nightcore, 8d, bass boost, lofi, remix, cover, karaoke, instrumental, mashup, tiktok — `_ALTERED_ASK`) count as asking without calling the AI, so a Groq timeout can't filter out a version you named.

Altered versions are only played when you ask for them: the main search checks every candidate with `_is_altered` ; matching a Spotify/Apple Music song to YouTube skips uploads whose title looks altered (unless the original's does); autoplay skips them by title. Only if *every* candidate is altered does the search fall back to the best one.

def _obvious_pick(ranked_top, asked, personal) Skips the AI when the answer is clear: the top result is the kind of version asked for (original vs altered, by title words) and either the user's history strongly points at it or it beats every rival of that kind by `AI_SKIP_MARGIN` (30). Most searches end here, using no Groq tokens.

async def _ai_pick_cached(query, cands, notes) `_ai_pick` remembered for 6 hours per search text, stored by song id so it still applies when the result list shifts.

async def _ai_pick(query, cands) One AI call when the answer isn't obvious (retried once if Groq rejects its own JSON). It gets the search text and the top `AI_PICK_CANDIDATES` (6) results (title, channel, length, views) and returns JSON: `wants_altered` (did the search explicitly ask for a sped up / slowed / remix / cover ... version; genre, artist, language or mood words don't count), `altered` (which results are non-original versions, including mashups, montagens and medleys) and `pick` (the result that is the song the user means). Returns None when Groq fails or times out (8 s); the search then uses the title rules.

The model is a reasoning model: its thinking counts toward `max_tokens` . All Groq calls use `_GROQ_ARGS` (400 tokens, low reasoning effort). Earlier versions allowed 3 tokens, so every answer came back empty and was read as "not altered", which is why altered versions used to slip through.

async def _groq_similar(seeds, count=5) Autoplay fallback: asks the model for `count` real songs similar to the seeds, one "Title - Artist" per line, and returns them as search queries. Errors return an empty list.

Source helpers and scoring

def _source_label(hint, query) Display name of the source (Spotify, SoundCloud, YouTube, Apple Music) from the URL, else from the `sp / sc / yt` hint.

def _is_local(track) True for Spotify local files and other non-streamable URIs; they are skipped in playlists.

def _fold(text) , **def _words(text)** Lower-case and strip accents ("Tântrico" → "tantrico"); split into words of 2+ characters. All matching is accent-insensitive.

def _track_score(track, query="", ref=None) Relevance of a search candidate. Each query word in the title +40, in the author +20. With a reference track (the Spotify match) each of its words +15 in the title or +5 in the author, and the length difference +60 (≤ 3 s), +25 (≤ 8 s) or -40 (> 20 s). "Official" +10; altered-version words -20 (`_ALTERED_QUICK`). Medleys (titles with two or more "/") -60; tracks over 10 minutes -30 unless the query asks for a mix/full/album/hour/live; mashups ("Song A x Song B") -50 and snippets ("best part") -30 unless the query has them.

async def _taste(guild_id, user_id) The requester's history summarised for searching: saved uploads, songs (title + artist), titles, artist play counts and search memory. Cached 60 s; dropped when they finish or skip a song, or reset.

def _personal(track, taste, query_norm) The personal score for one search result and a note for the AI (table in section 5).

async def _remembered_candidates(taste, query_norm, existing) Up to 3 uploads from the user's history that fit the search but aren't among the results, fetched by video id.

def _artist_key(author), def _song_id(title, author), def _query_norm(query) Normalised artist (no "- Topic"/"VEVO"), song identity and search text used by the history features.

async def _apply_normalize(player, on) Turns the LavaDSPX normalization filter (maxAmplitude 0.75, adaptive) on or off for a player. Filters stay on the player across songs, including the shutdown message.

def _popularity(meta) Popularity points from YouTube stats: $8 \times \log_{10}(\text{views} + 1) + 4 \times \log_{10}(\text{likes} + 1)$. About 99 for a billion-view hit, 48 for 50k views, 0 for an unwatched upload. Relevance still dominates: a song matching one fewer query word (-40) needs roughly 100× more views to win.

Embeds and messages

def _track_embed(track, action, source_label, queue_pos=0, search_path="") Now Playing / Added to Queue embed (title, artist, thumbnail, duration, source; search_path is kept for the logs but not shown). Autoplay songs show the source "Autoplay (Algo)".

def _now_playing_embed(track, source_label, search_path="") Shortcut for the Now Playing embed sent on track changes.

async def _send_status(channel, embed) Sends an embed and deletes it after 10 s (`_STATUS_DELETE_DELAY`).

async def _delete_after(msg, delay) Deletes a message after a delay, ignoring errors.

async def _set_vc_status(guild_id, channel_id, title) Sets the voice channel status to 🎵Playing - <title> (max 500 characters), or clears it with None. Needs the Set Voice Channel Status permission; failures are only logged.

Track bookkeeping

def _track_key(track) Stable key: identifier, else URI, else "title:author".

def _remember_alternates(sess, track, candidates, source_label, search_path) Stores the other search candidates as fallbacks for the chosen track in that session (no-op without a session, e.g. for autoplay AI searches).

def _track_meta(origin, yt_src) The algo record for a song: key (YouTube id when known, else URL), title, author, uri, yt_id.

Events to the controller

def _notify(guild_id) Queues a state event with the session's snapshot (or None when it ended).

async def _event_sender() Sends queued events to `.main.sock` one at a time, so they arrive in order. A lost event is harmless because the controller resyncs.

def _bg(fn, *args) Runs a blocking DB function in a thread in the background and logs errors.

yt-dlp resolver and playback

async def _ytdlp_run(video_url) Runs `python -m yt_dlp -g -f bestaudio/best ...` and returns the direct stream URL (nothing is downloaded). 20 s limit.

async def _ytdlp_url(video_url) Cached (1 h) and de-duplicated wrapper around `_ytdlp_run`.

async def _yt_meta(video_url) One `yt-dlp -j` call that returns a video's views, likes and direct stream URL. Cached (1 h) and de-duplicated; the stream URL is also put in the resolver cache, so the chosen track starts without a second yt-dlp run.

async def _rank_by_popularity(scored) Looks up `_yt_meta` for the `POP_CHECK` (4) most relevant candidates at once (skipping obviously altered titles unless you asked for one) (about 2 s on the Pi), adds

`_popularity` to their relevance, logs each line (relevance + popularity = total, views, likes) and returns all candidates re-sorted.

`async def _resolve(track)` Turns a track into something Lavalink can play. A YouTube video is tried three ways before giving up on it: the cached yt-dlp stream URL, a freshly fetched one (YouTube sometimes refuses a cached URL), then Lavalink's own YouTube plugin. Spotify/Apple Music tracks are matched to the best YouTube Music result first (never to an altered upload unless the original is one); non-YouTube sources play as they are; YouTube tracks are resolved with yt-dlp. Returns (playable, original track, YouTube track or None) or None.

`def _prefetch_next(sess)` Resolves the next queued YouTube track in the background.

`def _set_current(sess, origin, yt_src, requester)` Records the playing track's real title, its algo record and requester, and adds it to the history.

`def _after_start(sess)` After a track starts: sets the voice channel status, cancels the idle timer (or starts it when nobody is in the channel, so the bot never plays to an empty channel forever), prefetches the next track, and starts an autoplay refill when autoplay is on and the queue is empty.

`async def _play(sess, player, track, requester)` Plays a track, trying it and then its alternates (at most `MAX_RESOLVE_ATTEMPTS`, 2). Alternates are only other uploads of the same song (`_same_song`). When nothing resolves it hands the original to Lavalink anyway. Returns the track actually playing, which is what the reply, the Now Playing message and the channel status show.

`def _account(sess, played_ms=None, finished=False, skipped=False)` Feeds the outgoing track into its requester's saved songs: a listen when finished or played ≥ 30 s, a skip when skipped earlier, nothing otherwise. Writes happen in the background.

`def _enqueue(sess, item)` Adds a user's track ahead of any autoplay picks and returns its 1-based position.

`async def _advance(sess, player, announce=True)` Plays the next queued track, refilling from autoplay first when the queue is empty and autoplay is on. With nothing left it clears the voice channel status and starts the idle timer. Holds the session's `play_lock`, like the play commands, so only one song change runs at a time.

Autoplay

`def _same_song(a, b)` True when two results are uploads of the same song: same normalised title and either the same artist or a length within 8 s. Used to limit fallbacks, so a same-named song by someone else is never played in place of the one picked.

`def _norm_title(title)` Lower-cases a title and strips brackets and words like "official", "lyrics", "video", so different uploads of the same song compare equal.

`def _excluded_sets(sess)` Video ids and normalised titles already played in the session or queued.

`def _usable_rec(t, ids, titles)` True for a pick that isn't excluded, isn't an altered version and is 1–10 minutes long.

`async def _yt_id_for(seed)` The seed's YouTube id, looking it up on YouTube Music when unknown.

`async def _mix_for(seed)` Loads YouTube's radio mix for the seed and returns its tracks without the seed itself.

`def _weighted_pick(rows, k)` Picks up to k distinct saved songs at random, weighted by score $\times$ recency.

`async def _autoplay_seeds(sess)` Seeds for the next picks: weighted saved songs of `autoplay_user` in `algo` mode, else the most recent songs of the session (`autoplay_seed_count`).

`async def _mix_cached(sess, seed)` A seed's Mix, loaded once per session.

`def _autoplay_score(t, position, seed_weight, appearances, taste)` Score for one autoplay candidate, or None when the user skipped it in autoplay before (see "How picks are found"). `taste_w` is how much history counts: 0.6 for songs the user knows, `AUTOPLAY_NEW_TASTE_W` (0.15) for new ones.

async def _recommend(sess, count) Finds up to `count` picks. The pooled Mix songs are split into new-to-the-user and known (`_known_song`); each pick is new with probability `AUTOPLAY_NEW_SHARE` (0.7), else known (in algo mode a saved seed first, then known Mix songs). A missing kind is filled by the other; AI suggestions (batched, kept in `sess.ai_reserve`, new songs only) cover any remaining shortfall. Logs how many picks were new.

def _known_song(t, taste) True when the song is in the user's algo: the same upload, the same title and artist, or a re-upload whose title contains a saved song's title and artist.

async def _autoplay_refill(sess) Queues the picks (credited to the autoplay user or controller) if the queue is still empty and the session still exists.

def _kick_autoplay(sess) Starts a background refill unless one is running.

async def _autoplay_fill(sess) Starts (or joins) a refill and waits for it.

Idle and session lifecycle

def _start_idle_timer(sess), def _cancel_idle_timer(sess) Start (if not running) or cancel the session's idle countdown.

def _humans_in_vc(sess) True when someone who isn't a bot is in the session's voice channel.

async def _idle_countdown(sess) After `IDLE_TIMEOUT` (180 s) ends the session, unless music is actively playing (not auto-paused) with people in the channel.

async def _end_session(guild_id, disconnect=True) Removes the session, cancels its timers and autoplay, records the current track's listen, clears the voice channel status and leaves the channel (when `disconnect`), and notifies the controller.

async def _leave_voice(guild_id) Really leaves the voice channel: stop, disconnect, and if Discord still shows the bot in a channel, leave it directly and drop the voice client. Each step is separate, so a failing Lavalink call can't leave the bot sitting in the channel.

async def _fresh_vc(sess, new) Connects for a play command. A new session never reuses a player left over from an earlier one (a dead leftover holding an old paused track used to make new songs queue behind it and never start).

async def on_wavelink_track_start(payload) Lavalink says the audio really started (loading takes a moment after `play()`): restarts the session's playback clock (`_mark_position`) so lyrics line up with what people hear.

async def on_wavelink_track_stuck(payload) Lavalink reports a stream that stopped delivering audio: the song is skipped (not counted as the user's skip) and playback moves on.

async def _advance_safely(sess, player, announce=True) `_advance` for event handlers: on an error the session goes idle (and leaves after the idle timeout) instead of hanging. `_advance` itself skips songs Lavalink refuses to play and stops after `MAX_FAIL_STREAK` failures in a row.

async def _abandoned(sess, reply) After searching/connecting: if the session was stopped meanwhile (`/stop`, or a restart began), it doesn't play and leaves voice instead of leaving music running with no session.

def _not_found_msg(query, default) Spotify links that fail get an explanation (Spotify refusing lookups) instead of "Nothing found".

async def _session_watchdog() / async def _watchdog_pass() Every 30 s (and 10 s after start): resumes a **stalled queue** (songs waiting but nothing playing for two checks in a row, e.g. after a Lavalink restart; never while a song change holds `play_lock`, e.g. a slow YouTube lookup), leaves any voice channel the worker is in without a session (a "ghost" connection Discord kept after a restart), ends sessions whose voice connection is gone, and starts the idle timer for sessions with nothing to play, so a worker can't stay "busy" or sit in a channel forever.

def _claim(guild_id, channel_id, user_id) Returns the server's session (creating it with `user_id` as controller when there is none). Raises when the worker already plays in a different channel of that server.

async def _do_shutdown() After the `shutdown_graceful` reply: runs `_shutdown` and stops the event loop.

def _get_player(guild_id) The wavelink player (voice client) in a server, or None.

def _session_for_player(player) The session belonging to a player's server, or None.

Discord and Lavalink events

async def _enforce_whitelist(guild) Leaves a non-whitelisted server silently (the main bot posts the notice).

async def on_ready() Enforces the whitelist; on the **first** ready only, starts the event sender, the Lavalink connection and the IPC server (Discord reconnects fire `on_ready` again, and these must not start twice).

async def on_guild_join(guild) Enforces the whitelist.

async def on_wavelink_node_ready(payload), **async def on_wavelink_node_disconnected(payload)** Track and log the Lavalink connection (wavelink reconnects by itself).

async def on_wavelink_track_end(payload) For the session of that server: a `finished` track resets the fail streak and counts as a listen. `loadFailed`, `replaced`, `stopped` and `cleanup` are ignored (already handled elsewhere), as is anything during a skip. Otherwise it re-queues the track when loop is on and advances.

async def on_wavelink_track_exception(payload) Logs the error and counts the failure. After `MAX_FAIL_STREAK` (3) in a row it posts "Playback failed", clears the queue, turns autoplay off, stops and starts the idle timer. Otherwise it tries the failed track's next alternate, or advances.

async def on_voice_state_update(member, before, after) - The worker itself left voice (kicked or disconnected by someone) → the session ends, so the worker is free again. - The worker was dragged to another channel → the session follows it, and the "🎵Playing - title" status moves to the new channel. - The last person left the session's channel → playback pauses and the idle timer starts. - Someone came back → playback resumes and the idle timer stops.

Connections

async def _connect_lavalink() Connects to Lavalink, retrying forever (5 s more per attempt, at most 30 s).

async def _connect_vc(sess) Returns a player connected to the session's channel: reuses, moves or connects with `VC_RETRIES` attempts of `VC_TIMEOUT` each. Raises a readable error when the worker isn't in the server or the channel is invalid.

Search

async def _sp_lookup(query) Spotify search (`spsearch:`) for clean title/artist/length. Of the top 5 results it takes the one containing the most of the query's words, and only if it contains at least half of them (`SP_MIN_COVERAGE`); otherwise None. Spotify's first hit can be unrelated, and Spotify errors also give None.

async def _am_load(url) Loads an Apple Music track, album or playlist through LavaSrc.

async def _yt(query, count=8), **async def _sc(query, count=5)** YouTube Music / SoundCloud search through Lavalink.

async def _first_clean(candidates) First candidate the Groq filter doesn't consider altered.

async def _search(query, source, sess) Finds the track for a query and returns (`track`, `source label`, `found-via`). Apple Music and other URLs load directly. `sc` searches SoundCloud and prefers a non-altered result. `sp` (default) runs the Spotify lookup, the altered-version check and a YouTube Music search for the exact query at the same time; if Spotify gave a matching song, YouTube Music is also searched with its clean title/artist and both result lists are merged. `yt` skips Spotify. Candidates are ranked by relevance plus popularity (`_rank_by_popularity`), then `_ai_pick` decides. Its answer is checked: unless an altered version was asked for, results the AI or obvious title words mark as altered are never picked; and the pick must match the search about as well as the best result (relevance within 40), so a slowed version of a *different* song can't win. If the AI fails, obvious title words decide instead. Remaining allowed results become the fallbacks. Only if

every result is altered does it try preferred lyric channels, then the best-ranked result. An explicitly requested altered version skips the filter. Alternates are stored on the session.

async def _search_playlist(query, source) Loads a playlist (Apple Music, URL, SoundCloud search or Spotify search) and returns (playable tracks, name, source label, skipped local files).

IPC server and shutdown

def _track_reply(track, embed_type, lbl, path, **extra) The reply fields the controller turns into a track embed.

async def _handle_connection(reader, writer) Handles one IPC connection: reads one JSON line, runs the operation (section 9) and writes one JSON reply. Operations that can start playback go through `_claim` and `_start_op`; if a new session ends up playing nothing (search failed, connect failed), it is ended again so the worker is free. Any exception becomes an error reply.

async def _start_op(sess, op, cmd, uid, reply) The `search_and_play / queue_track`, `search_and_playlist` and `autoplay_start` operations.

async def _ipc_server() Starts the Unix socket server on `.worker<N>.sock`.

async def _shutdown() Closes the Discord client and deletes the socket file.

8. db.py and appsettings.py reference

db.py

All functions are blocking SQLite calls (WAL mode, 10 s busy timeout), safe to use from the controller and all workers at once. The worker calls them in a background thread.

def _connect(path) Opens a SQLite file (creating its folder) with row access by name and WAL mode.

def _server_con() Opens `database/server/servers.db` and creates or upgrades the `guild_settings` table (`guild_id`, `cc_id`, `vcw` JSON, `eph` JSON, `src`, `updated_at`).

def get_guild(guild_id) A server's settings as `{cc_id, vcw, eph, src, rtc_region}`; defaults when the server has none. The `rtc_region` column is added to an existing `servers.db` automatically.

def save_guild(guild_id, cfg) Inserts or updates a server's settings.

def guild_exists(guild_id) True when the server has stored settings.

def _algo_con(guild_id) Opens `database/algo/<guild_id>.db` and creates the `user_tracks` table (`user_id`, `track_key`, `title`, `author`, `uri`, `yt_id`, `plays`, `skips`, `last_played`, `added_at`; one row per user and song).

def record_play(guild_id, user_id, meta, max_kb) Adds a listen (inserting the song with 1 play, or +1 play and refreshing its details), then evicts.

def _evict(con, user_id, keep_key, max_kb) Removes the user's lowest-score songs (oldest play breaks ties) until they are within the storage cap, never removing `keep_key`. There is no song-count limit.

def record_skip(guild_id, user_id, key) +1 skip on a saved song; does nothing for unsaved songs.

def user_tracks(guild_id, user_id, limit=100000) The user's saved songs, best score first, with a `score` field.

def record_query(guild_id, user_id, query, meta, listened, max_kb) Search memory: +1 listen or +1 skip for (user, search text, song). Capped at a quarter of `max_kb`, least recently used first.

def user_queries(guild_id, user_id), def user_query_count(guild_id, user_id) A user's search memory / its size.

def remove_user_tracks(guild_id, user_id, keys) Algo menu "Remove selected": deletes those songs (and the user's search memory for them) for one user only.

def reset_user(guild_id, user_id) /reset-algo: deletes one user's saved songs and search memory on one server; returns the counts.

def user_stats(guild_id, user_id) (song count, approximate bytes) for a user.

def lyrics_get(cache_key) / def lyrics_put(cache_key, provider, duration_ms, lines) The lyrics cache in `database/lyrics.db`: the chosen synced lines (JSON) with their provider and the source's duration, or `lines` NULL for "no synced lyrics anywhere".

appsettings.py

def get() Returns `server_settings.json` merged over the defaults, re-reading the file only when it changed. A broken file keeps the previous values and logs the error.

def require_startup() Used at start-up by the controller and workers. Exits with a clear message when the file is missing, has fewer than 2 tokens or no Lavalink URI.

def is_dev(user_id) True when the user id is in `dev_ids`.

def is_whitelisted(guild_id) True when the whitelist is off or the server is listed.

def get_int(name) A numeric setting (`algo_max_kb` , `autoplay_seed_count`). An invalid value (null, text) is logged once and the default is used, instead of raising during a song change.

9. IPC operations and state events

Controller → worker

The controller sends one JSON object per connection to `.worker<N>.sock`, with `op` plus `guild_id`, `channel_id`, `text_channel_id` and `user_id` where relevant. The worker answers with one JSON line: `status` (`ok` or `error`), `message`, and extra fields. Replies that change a session include `session` (its snapshot).

op	What the worker does	Reply
ping	Reports its health (and its session in <code>guild_id</code> , if given).	pong with <code>discord_ok</code> , <code>discord_ms</code> , <code>lavalink_ok</code> , <code>lavalink_uri</code> , <code>sessions</code> (count), <code>session</code> , <code>queue_len</code>
sync	Reports its full state.	synced with <code>user_id</code> , <code>guilds</code> , <code>lavalink_ok</code> , <code>sessions</code> (server id → snapshot)
search_and_play, queue_track	Claims the session, searches, connects, then plays or queues (ahead of autoplay picks).	playing or queued with <code>title</code> , <code>author</code> , <code>uri</code> , <code>artwork</code> , <code>duration</code> , <code>source_label</code> , <code>search_path</code> , <code>queue_pos</code> , <code>session</code>
search_and_playlist	Loads the playlist; plays the first and queues the rest, or queues all.	playing_playlist or queued_playlist with <code>pl_name</code> , <code>count</code> , <code>title</code> , <code>artwork</code> , <code>source_label</code> , <code>skipped</code> , <code>session</code>
autoplay_start	Checks the user has saved songs, turns on autoplay from them; starts playing, or switches an already-playing session.	playing (as above) or autoplay_on
stop	Ends the session (records the listen, clears the status, leaves).	stopped
pause_resume	Pauses or resumes.	paused or resumed
skip	Records a listen (≥ 30 s played) or a quick skip, stops the track and advances (refilling autoplay if needed).	skipped
backward	Seeks to the start.	restarted
seek	Moves by <code>delta_ms</code> , clamped to the track.	seeked with <code>delta_ms</code>
remove_from_queue	Deletes the song with <code>qid</code> . Error if it already played or was removed.	removed with <code>title</code> , <code>queue_remaining</code>
get_queue	Returns the current track and the queue: each entry has <code>title</code> , <code>author</code> , <code>index</code> (position), <code>qid</code> and an autoplay flag.	queue with <code>current</code> , <code>queue</code> , <code>muted</code> , <code>loop</code> , <code>autoplay</code> , <code>normalize</code> , <code>paused</code> , <code>lyrics</code> , <code>session</code>
toggle_mute	Hides or shows now-playing messages.	muted or unmuted
toggle_loop	Repeats the current track or not.	loop_on or loop_off
toggle_autoplay	Turns autoplay on (needs played history unless in algo mode; starts playing if idle) or off (drops queued picks).	autoplay_on or autoplay_off, <code>session</code>
set_mode	Sets mode to <code>me</code> or <code>all</code> .	mode_me or mode_all, <code>session</code>

op	What the worker does	Reply
toggle_normalize	Turns the normalization filter on or off.	normalize_on or normalize_off
set_no_eph	/control no_eph : makes the session's replies public (on: true ; also turns on now-playing posts) or not.	no_eph_on or no_eph_off with session
toggle_lyrics	Turns lyrics on or off for the session (on sets it, without on it toggles). Lyrics are private when the playback was started privately.	lyrics_on or lyrics_off with lyrics, private
forget_user	After /reset-algo : drops the user's cached history and doesn't re-save the song playing now.	forgotten
jump_to	Records a listen or quick skip for the current track, drops the queue up to the song with qid and plays it. Error if it already played or was removed.	jumped with title, author, queue_remaining
shutdown_graceful	Ends every session, replies, then exits.	shutdown_graceful
drain	Graceful restart: blocks new playback; each session finishes its song, plays the message at message_url and leaves.	draining with remaining
drain_force	Restart timeout: cuts songs still playing and plays the message now.	forced with remaining

Session operations on a server where the worker has no session reply "Nothing playing.". Any other op returns "Unknown op".

Worker → controller

Sent to .main.sock , one JSON line per connection, in order:

```
{ "op": "state", "index": 2, "guild_id": 123, "session": { "channel_id": 456, "controller_id": 789, "mode": "me", "autoplay": false, "text_channel_id": 111, "sid": 5, "no_eph": false } }
```

session is null when the session ended. Sent when a session is created or ends, and on autoplay, mode and channel changes.

When something a panel shows changes (next song picked or started, queue changed), the worker sends { "op": "panel", "index": 2, "guild_id": 123 } so open panels redraw right away. A playlist being added sends one event, not one per song.

For private lyrics the worker also sends:

```
{ "op": "lyrics", "index": 2, "guild_id": 123, "song": 7, "seq": 41, "content": null, "embed": { "title": "Lyrics - Song", "description": "..."}, "delete": false }
```

song numbers the song's message and seq orders updates. The update is an embed (content is supported too). delete: true means the song ended and its message goes away.

10. Tunable constants and files

Name	File	Value	Meaning
<code>SERVICE_NAME</code>	main.py	<code>SERVER_musicbots</code>	systemd unit used by <code>restart full / shutdown full</code> .
<code>DRAIN_TIMEOUT</code>	main.py	300 s	Graceful restart: longest wait for current songs before they are cut.
<code>DRAIN_MESSAGE_MAX</code>	main.py	130 s	Graceful restart: extra wait for the shutdown messages after the timeout.
<code>SHUTDOWN_MESSAGE_MAX</code>	worker.py	120 s	A worker leaves this long after starting the message, even if it never reports finishing.
<code>PENDING_TTL</code>	main.py	90 s	How long a <code>/play</code> reservation holds a worker without confirmation.
<code>PLAY_TIMEOUT</code>	main.py	90 s	How long <code>/play</code> , <code>/playlist</code> and <code>/autoplay</code> wait for the worker (search + voice connect + stream).
<code>IPC_LINE_LIMIT</code>	main.py	8 MB	Longest worker reply accepted (big playlist queues).
<code>HUNG_AFTER</code>	main.py	4	Failed 30-second health checks in a row before a running-but-frozen worker is killed and restarted.
<code>PANEL_REFRESH</code>	main.py	4 s	How often an open <code>/control</code> panel checks for changes (worker <code>panel</code> events redraw it right away).
<code>PANEL_LIFETIME</code>	main.py	14 min	A panel's life: Discord allows editing it for 15 minutes. Public panels are deleted then, private ones marked expired.
<code>PANEL_END_NOTE</code>	main.py	5 s	How long an ended panel shows "Stopped" / "Playback ended" before it's deleted.
<code>PAGE_SIZE</code>	main.py	25	Queue entries per dropdown page (Discord limit).
ControlView timeout	main.py	900 s	Control panel buttons stop working after 15 minutes.
SettingsView timeout	main.py	600 s	Settings panel stops working after 10 minutes.
<code>VC_TIMEOUT</code> , <code>VC_RETRIES</code>	worker.py	60 s, 3	Time and attempts for joining or moving to a voice channel.
<code>IDLE_TIMEOUT</code>	worker.py	180 s	Idle time before the worker leaves.
<code>YTDLP_TIMEOUT</code>	worker.py	20 s	Maximum time for one yt-dlp lookup.
<code>YTDLP_NICE</code>	worker.py	10	yt-dlp runs at this lower CPU priority, so lookups never take CPU from Lavalink's audio.
<code>PRELOAD_BEFORE_MS</code>	worker.py	30 s	This long before a song ends, the next song's stream is fetched fresh and loaded.
<code>PRELOAD_TTL</code>	worker.py	120 s	How long a preloaded stream is trusted (YouTube stream links go stale).

Name	File	Value	Meaning
URL_CACHE_TTL	worker.py	3600 s	How long a resolved stream URL is reused.
MAX_RESOLVE_ATTEMPTS	worker.py	2	Candidates tried per play before handing the original to Lavalink.
MAX_FAIL_STREAK	worker.py	3	Failed tracks in a row before the worker stops and reports.
POP_CHECK	worker.py	3	Candidates whose YouTube views/likes are looked up per search (each is a YouTube request).
META_CACHE_TTL	worker.py	24 h	How long views/likes are reused (fewer YouTube requests).
POP_VIEW_W, POP_LIKE_W	worker.py	8, 4	Popularity points per 10× views / likes.
SP_MIN_COVERAGE	worker.py	0.5	Share of the query's words a Spotify match must contain to be used.
PLAY_COUNT_MS	worker.py	30 s	Listening time for a song to count as a listen.
HISTORY_LEN	worker.py	25	Played tracks remembered per session.
AUTOPLAY_BATCH	worker.py	5	Autoplay picks queued per refill.
AUTOPLAY_NEW_SHARE	worker.py	0.7	Chance each autoplay pick is a song new to the user (else one they know).
AUTOPLAY_NEW_TASTE_W	worker.py	0.15	How much the artists you play count when ranking new songs (about half of new picks end up by them).
AUTOPLAY_MIX_DEPTH	worker.py	30	Songs used from each seed's YouTube Mix.
_STATUS_DELETE_DELAY	worker.py	10 s	Lifetime of the now-playing messages.
AI_PICK_CANDIDATES	worker.py	6	Search results shown to the AI when it's needed.
AI_SKIP_MARGIN	worker.py	30	Lead that makes a result a clear winner (no AI call).
AI_CACHE_TTL	worker.py	6 h	How long an AI decision is reused for the same search.
AUTOPLAY_PER_ARTIST	worker.py	2	Max songs by one artist per autoplay refill.
PERSONAL_STRONG	worker.py	40	Personal score at which your history overrides the AI pick.
PERSONAL_HALF_LIFE_DAYS	worker.py	30	How fast personal boosts fade without listening.
NORMALIZE_SETTINGS	worker.py	maxAmplitude 0.75, adaptive	Normalize filter settings.
LYRICS_TICK	worker.py	0.25 s	How often the lyrics loop checks the playback position.
LYRICS_EDIT_LEAD	worker.py	0.4 s	Lines are sent this early, the time an edit takes to show in Discord. <code>lyrics_delay</code> in <code>server_settings.json</code> adjusts on top (-0.2 = 0.2 s earlier).
PAIR_MAX_GAP_MS	lyrics.py	7 s	With <code>lyric_safe</code> , two lyric lines share a display line only if the second starts within this.

Name	File	Value	Meaning
LYRICS_MIN_GAP	worker.py	1.2 s	At most one lyrics edit this often; faster lines skip to the latest.
PROVIDER_TIMEOUT	lyrics.py	8 s	Per lyrics provider; one that's slower is skipped.
MAX_LENGTH_DIFF_MS	lyrics.py	3 s	Lyrics whose source duration is further off are another version and aren't used.
NONE_CACHE_TTL , FOUND_CACHE_TTL	lyrics.py	3 days, 30 days	How long "no synced lyrics" and found lyrics are cached.
GAP_MS	lyrics.py	8 s	When the next line is 12 s or more away (or after the last line), a block starts this long after the line. It's shown as next before it starts.
_GROQ_ARGS	worker.py	400 tokens, low reasoning	Settings for every Groq call (the AI pick allows 1200 tokens).

File	Created by	Purpose
server_settings.json	you	Tokens, Lavalink, devs, whitelist, algo and autoplay limits. Secret, git-ignored.
server_settings.json.example	repo	Template for the above.
.env	you	GROQ_API_KEY . Secret, git-ignored.
database/server/servers.db	controller	Per-server settings from /settings .
database/algo/<server id>.db	workers	Every user's saved songs for that server.
database/lyrics.db	workers	Lyrics cache (see section 13).
.worker<N>.sock	worker	Socket the controller uses to talk to worker N.
.main.sock	controller	Socket the workers send state events to.
shutdown.mp3	you	Message played before a restart (optional).
launcher.py	repo	systemd entry point that makes restart/stop return immediately (section 11).
.bot.pid	controller	PID of the running bot, so a new launcher can wait for an old one still finishing.
settings.json.migrated	controller	The old single-server settings after their one-time import.

To reset a server's settings, delete its row from `servers.db` (or the whole file to reset every server). To forget everyone's saved songs on a server, delete `database/algo/<server id>.db`.

11. Graceful restart

`sudo systemctl restart SERVER_musicbots` and `sudo systemctl stop SERVER_musicbots` (and `restart full / shutdown full`) **return immediately** and don't cut the music. The waiting happens in the background:

1. systemd runs **launcher.py**, which starts `main.py --server` as its child. On restart/stop systemd signals **only the launcher** (`KillMode=process`).
2. The launcher sends `{"op": "shutdown"}` to the bot over `.main.sock`, waits for its **ACK**, and exits. systemd sees the stop as done, so `systemctl` returns. (No ACK within 10 s → the launcher sends the bot SIGTERM, which it handles the same way.)
3. The bot blocks commands that start playback ("restarting — try again in a few minutes"; `/stop` and `/control` still work) and sends `drain` to every worker. A draining worker refuses new playback ("hard block"), even after its own song finished.
4. Each worker, per server: - a song is playing → it finishes **that song**; the rest of the queue/playlist and autoplay are dropped; - paused or between songs → the message plays right away; - nobody in the voice channel → it leaves without a message.
5. After the song, the worker plays **shutdown.mp3** (next to `main.py`) and leaves when it ends (at most 2 minutes). Without the file, it just leaves.
6. The bot waits until **every** worker is done, or `DRAIN_TIMEOUT` (5 minutes). At the timeout, songs still playing are cut and the message plays immediately (`drain_force`); then it waits for those messages and exits.
7. On a restart, systemd has already started a **new launcher**. It sees the old bot still running (its PID is in `.bot.pid`), logs "Previous bot ... is still finishing songs — waiting", and starts the new bot once the old one exits (after at most 9 minutes it kills the old one).

A second stop request while the bot waits (another `systemctl restart / stop`) skips the wait and goes straight to step 6. If nothing is playing, it restarts in a few seconds. If the bot crashes, the launcher exits with the same code, so `Restart=on-failure` restarts it as before.

The message is served to Lavalink over a private `http://127.0.0.1:<random port>/shutdown.mp3` by the bot, so Lavalink must run on the same machine (its `http` source is enabled; its `local` file source is not needed).

Service settings (in `/etc/systemd/system/SERVER_musicbots.service`):

```
ExecStart=/home/jared/servers/musicbot_updated/venv/bin/python launcher.py
KillMode=process
TimeoutStopSec=30
Restart=on-failure
```

`KillMode=process` is what lets the bot outlive the launcher during its graceful shutdown; the bot stops its workers itself.

launcher.py reference

def main() Waits for a previous bot that's still finishing (from `.bot.pid`, up to `OLD_BOT_WAIT`, 540 s), starts `main.py --server`, and waits for it. Exits with the bot's code if it stops by itself.

def _previous_bot() PID of a still-running bot from `.bot.pid` (checked against `/proc/<pid>/cmdline`), or None.

def _ask_bot_to_stop(pid) Sends the shutdown request and waits for the ACK (`ACK_TIMEOUT`, 10 s); falls back to SIGTERM.

main.py: def _request_stop() Starts the graceful shutdown; a second call cuts songs right away. Used by the launcher's shutdown request and by SIGTERM/SIGINT.

12. Deployment and service

The service file is `/etc/systemd/system/SERVER_musicbots.service`. It runs `venv/bin/python launcher.py` (which starts `main.py --server`) from `/home/jared/servers/musicbot_updated`, after `SERVER_lavalink`. A backup of the version that pointed at the old folder is at `~/servers/SERVER_musicbots.service.bak`.

Fresh install:

```
python3 -m venv venv
```

```
venv/bin/pip install -r requirements.txt
```

Then copy `server_settings.json.example` to `server_settings.json`, fill it in, create `.env` with `GRQQ_API_KEY=...`, and start the service:

```
sudo systemctl start SERVER_musicbots
```

Follow the logs (controller and every worker, prefixed `[Worker N]`):

```
journalctl -u SERVER_musicbots -f
```

Notes:

- Don't run a second copy (for testing) while the service is running: they share the same bot tokens.
- Slash commands are registered per server on every start, so new or changed commands show up right away (Ctrl+R in Discord if an open client still shows the old list).
- Spotify's Web API requires the owner of the Spotify developer app (configured in Lavalink's `application.yml`) to have an active Premium subscription. Without it, Spotify searches and `open.spotify.com` links fail; searches then fall back to YouTube Music only.
- The Message Content Intent is a privileged intent; beyond 100 servers Discord requires the bot to be verified for it.

13. Lyrics

Live **synced** lyrics for the song that's playing, turned on per session with  **Lyrics** in `/control` (there's no server default). Bots can't stream video, so lyrics are a message that's edited as the song plays:

- An embed titled **Lyrics - <song title>**. Its description is a code block with three blocks, previous, **current** (each row marked ) and next, divided by solid lines (—). The divider's length stays the same for the whole song: its longest line, capped at 30 characters so it doesn't wrap on a phone. The layout depends on `lyric_safe` in `server_settings.json`, which is read when each song's lyrics start:
- **true (default): two lyric lines per block**, each on its own row. A message shows 6 lyric lines and changes about half as often (40–49% fewer edits on real songs). The pairs are fixed: lines 1+2, 3+4 and so on. A music break () stays on its own block, and two lines more than 7 s apart aren't paired, so the second never shows long before it's sung.
- **false: one lyric line per block** (more edits).
- **Music breaks** () are blocks in the timeline: the ones the lyrics mark, and a  added 8 s after a line when the next line is 12 s or more away (and after the last line). So a break always shows up as the *next* block first, then becomes current. It never suddenly replaces the line being shown.
- **One message per song**, deleted when the song ends (or is skipped or stopped, or lyrics are turned off). The next song gets a new one.
- **Privacy follows the playback**. If playback was started with a private command, lyrics are an ephemeral message only the person who pressed  sees. Discord allows that for 15 minutes after the click, so it then says to press  again. Otherwise it's a normal message in the session's text channel, posted by the worker.
- **Only synced lyrics**. If no provider has synced lyrics for the song, the message just says "Synced lyrics not available". Plain unsynced text is never shown. If the lookup takes more than a second, the message first says "Looking up synced lyrics...".
- **Rate limits**: a message is edited only when the line changes, at most every 1.2 s (`LYRICS_MIN_GAP`). When lines change faster, it skips to the latest instead of queueing edits. Nothing is edited while paused.

Where lyrics come from

`lyrics.py` asks three providers at the same time. One that fails or takes longer than `PROVIDER_TIMEOUT` (8 s) is skipped:

Provider	How it matches
YouTube Music (<code>ytmusicapi</code>)	The lyrics of the exact video being played, when it's a song upload. For a re-upload or lyric video, the song upload of the same recording (same title and artist, within 3 s of the same length).
LRCLib	Title, artist and duration (<code>/api/get</code> , then <code>/api/search</code>).
NetEase	Search by title and artist, then the song with matching title and duration. Its credit lines (作词/作曲...) are removed.

Re-uploads often put the real artist in the title ("Nakama, Mc Staff - MENTE MÁ" uploaded by "void"), so that split is searched too.

Choosing: a source whose duration is more than 3 s off the playing upload is a different version (slowed, sped up, a music-video intro) and is rejected, as is one that isn't really synced or runs past the end of the song. When two or more remain, the score favours, in order: the exact video (YT Music), the closest duration, then more timed lines.

Musixmatch isn't used: its synced lyrics are only available through its desktop app's internal API, which its terms forbid (YT Music's lyrics largely come from Musixmatch anyway). Genius isn't used because it has no synced lyrics.

Cache: results are kept in `database/lyrics.db`. Found lyrics last 30 days, and "no synced lyrics" lasts 3 days (only when every provider answered). When lyrics are on, the next song in the queue is looked up while the current one plays, so its lyrics usually appear at once.

Timing: wavelink only learns the playback position from Lavalink's updates (every 5 s), so right after a song starts, a seek, or a resume, its position is stale. The worker keeps its own clock (`_mark_position` / `_play_position`), restarted when Lavalink reports that the track really started, and uses Lavalink's position again once a newer update arrives. Lines are sent early to cover the edit delay: 0.4 s (`LYRICS_EDIT_LEAD`) minus `lyrics_delay` from `server_settings.json` (default -0.2, so 0.6 s in total). Change `lyrics_delay` if lines feel late (more negative) or early (towards positive).

Coverage at launch, checked against the saved songs on the main server: 51 of 69 had synced lyrics (48 from YouTube Music, 3 from LRCLib). Most of the rest are instrumental.

lyrics.py reference

async def find(title, artist, length_ms, yt_id=None) Synced lyrics for the playing upload (`Lyrics` , or `None`), plus a short report for the log (provider names and counts only, never lyric text). Uses the cache, otherwise asks the providers. Concurrent lookups of the same song share one request.

def parse_lrc(text) LRC to `[(start_ms, text)]`. Handles several timestamps on one line, `[offset:]`, `[mm:ss.xx]` and `[mm:ss:xx]`, and word timestamps. Metadata and credit lines are dropped, and empty lines become `.`.

def clean_title(title, artist) / def clean_artist(author) The names lyrics sites use: no "(Official Video)", "feat." or "Artist - " prefix, and only the first artist. Version words like "(Slowed)" are kept.

def state_at(lyrics, pos_ms) / def render(state) The (previous, current, next) blocks at a position, with breaks added by `_with_breaks` (merged=True for `lyric_safe`: up to two lyric lines each, from `Lyrics.layouts`, built by `_group`), and their code-block text with `—` dividers (`Lyrics.width` long).

worker.py: lyrics

async def _lyrics_song(sess, player, track) One song's lyrics. It looks them up, then every `LYRICS_TICK` finds the display line at `_play_position` plus the lead (`LYRICS_EDIT_LEAD` - `lyrics_delay`), and edits only when it changed. It ends when the song ends, the session ends or lyrics are turned off, and its `finally` deletes the message.

class _LyricsOut Where one song's lyrics go: a message in the session's text channel (public playback), or `Lyrics` events to the controller (private playback). `_lyrics_message(title, body, safe)` builds the embed.

def _start_lyrics(sess) / def _stop_lyrics(sess) Start lyrics for the song playing now (called when a song starts and when  is turned on) and prefetch the next song's lyrics. Stopping cancels the task, which deletes the message.

def _mark_position(sess, pos_ms) / def _play_position(sess, player) / async def _set_paused(sess, player, paused) The playback clock described above. Pause, resume, seek and the start of each song set a mark.

textnorm.py

`fold`, `words`, `artist_key` and `norm_title`: the text normalisation shared by search, autoplay and lyrics (accents removed, "- Topic"/"VEVO" dropped from artist names, bracketed parts and "official video" dropped from titles).

14. Playback smoothness

Measured on the server (Raspberry Pi 5, 4 cores, wired gigabit, idle around 49 °C, no throttling): the hardware and network aren't the bottleneck, and the bots already connect to Hong Kong voice servers (the closest region) automatically. What reduces stutter and gaps:

- **Next song preloaded:** 30 s before a song ends (`PRELOAD_BEFORE_MS`), the worker fetches a **fresh** stream link for the next song and loads it in Lavalink, so the next song starts at once (`_preload_next`). Before, the link was fetched when the previous song started; YouTube often refused it minutes later (20 times in 3 days), and each refusal cost 2–3 s of silence while a new one was fetched. Not done in loop mode or for live streams. A preload older than `PRELOAD_TTL` (2 min) isn't used.
- **yt-dlp at lower CPU priority** (`YTDLP_NICE` , nice 10): searches and stream lookups run on the same Pi as Lavalink; they now always yield to the audio.
- **Bigger Lavalink audio buffer:** `frameBufferDurationMs: 10000` in `~/servers/lavalink/application.yml` (was 5000). Lavalink keeps up to 10 s of decoded audio ready, which rides out short YouTube hiccups. The side effect is that **Normalize** can take up to 10 s to be heard after toggling it. Changing it needs a Lavalink restart. The backup of the previous file is `application.yml.bak-2026-10-10b`.
- **Voice region per server** (`/settings` →  Voice region). Options:
 - **Don't change** (default): channels keep their own region.
 - **Automatic:** Discord picks.
 - **A region** (Hong Kong, Singapore, Japan, ...).

The region is set on the voice channel just before a bot joins it (`_apply_region`). The worker bots need **Manage Channels** for this, and people already in the channel reconnect for a moment. Pick the region nearest to most listeners. It's only worth changing if the automatic choice is far from them. Switching back to "Don't change" doesn't revert channels already changed; change those in the channel's settings.

If audio still stutters, Lavalink's own counters show whether frames are being lost on the server side (`GET /v4/stats` → `frameStats: sent, nulled, deficit` per minute while something plays).